

Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

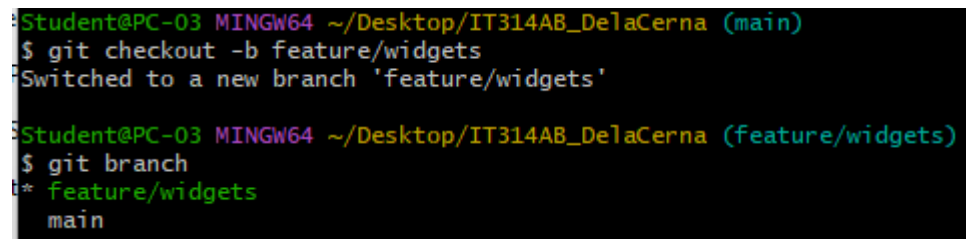
(I know what AI-generated code looks like, don't even think about it :>)

🚩 Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`



```
Student@PC-03 MINGW64 ~/Desktop/IT314AB_DelaCerna (main)
$ git checkout -b feature/widgets
Switched to a new branch 'feature/widgets'

Student@PC-03 MINGW64 ~/Desktop/IT314AB_DelaCerna (feature/widgets)
$ git branch
* feature/widgets
main
```

🚩 Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

Text(

'Chiara Marie C. Canque\n'

'BSIT - 3\n'

'My First Flutter Application\n'

'August 11, 2026',

)

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

Text → Application Title

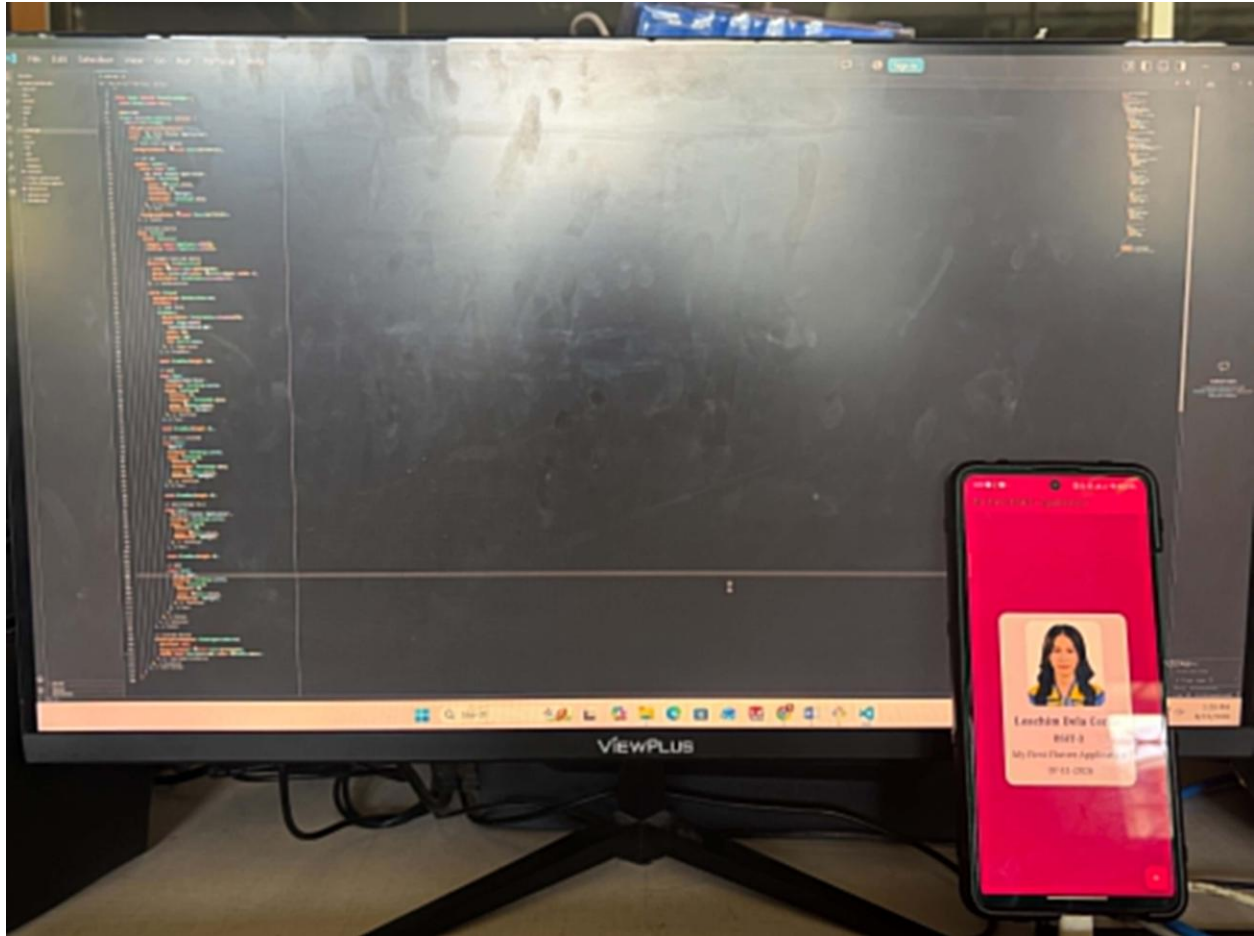
Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets
- Your application

```
main.dart
1  class MyApp extends StatelessWidget {
2    const MyApp({super.key});
3
4    @override
5    Widget build(BuildContext context) {
6      return MaterialApp(
7        debugShowCheckedBanner: false,
8        title: 'My First Flutter Application',
9        home: Scaffold(
10          // DARK PINK BACKGROUND
11          backgroundColor: const Color(0xFFD1457),
12
13          // APP BAR
14          appBar: AppBar(
15            title: const Text(
16              'My First Flutter Application',
17            ),
18            style: TextStyle(
19              color: Colors.black,
20              fontSize: 30,
21              fontFamily: 'Georgia',
22              fontWeight: FontWeight.bold,
23            ), // Text
24            backgroundColor: const Color(0xFFD1457),
25          ), // AppBar
26
27          // CONTAINER CONTENT
28          body: Container(
29            margin: const EdgeInsets.all(20),
30            padding: const EdgeInsets.all(20),
31
32            // ROUNDED CARD AND BORDER
33            decoration: BoxDecoration(
34              color: const Color(0xFFD1457),
35              border: Border.all(color: Colors.black, width: 1),
36              borderRadius: BorderRadius.circular(15),
37            ), // BoxDecoration
38
39            child: Column(
40              mainAxisAlignment: MainAxisAlignment.min,
41              children: [
42                // VIEW PICCO
43                ClipRect(
44                  borderRadius: BorderRadius.circular(20),
45                  child: Image.asset(
46                    'assets/GoogleCar.png',
47                    width: 200,
48                    height: 200,
49                    fit: BoxFit.cover,
50                  ), // Image.asset
51                ), // ClipRect
52
53                const SizedBox(height: 20),
54
55                // NAME
56                const Text(
57                  'Launch Data Car',
58                  textAlign: TextAlign.center,
59                  style: TextStyle(
60                    fontSize: 20,
61                    fontWeight: FontWeight.bold,
62                    color: Colors.black,
63                    fontFamily: 'Georgia',
64                  ), // Text
65                ), // Text
66
67                const SizedBox(height: 10),
68
69                // COURSE & SECTION
70                const Text(
71                  'Flutter 2',
72                  textAlign: TextAlign.center,
73                  style: TextStyle(
74                    fontSize: 22,
75                    fontWeight: FontWeight.bold,
76                    color: Colors.black,
77                    fontFamily: 'Georgia',
78                  ), // Text
79                ), // Text
80
81                const SizedBox(height: 10),
82
83                // APPLICATION TITLE
84                const Text(
85                  'My First Flutter Application',
86                  textAlign: TextAlign.center,
87                  style: TextStyle(
88                    fontSize: 24,
89                    color: Colors.black,
90                    fontFamily: 'Georgia',
91                  ), // Text
92                ), // Text
93
94                const SizedBox(height: 10),
95
96                // DATE
97                const Text(
98                  '2023-11-20',
99                  textAlign: TextAlign.center,
100                 style: TextStyle(
101                   fontSize: 20,
102                   color: Colors.black,
103                   fontFamily: 'Georgia',
104                 ), // Text
105               ), // Text
106
107               ], // Column
108             ), // Container
109
110             // FLOATING BUTTON
111             floatingActionButton: FloatingActionButton(
112               onPressed: null,
113               backgroundColor: const Color(0xFFD1457),
114               child: const Icon(Icons.add, color: Colors.white),
115             ), // FloatingActionButton
116           ), // Scaffold
117         ), // MaterialApp
118       );
119     }
120   }
121 }
```



🚩 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

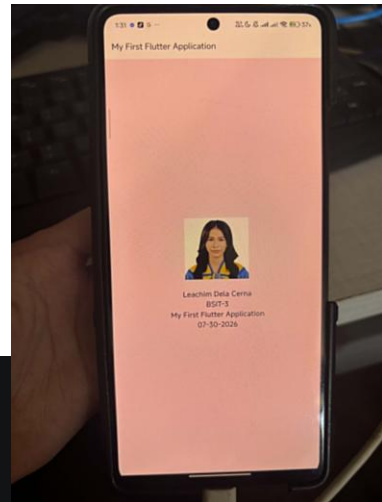
Course & Section

Application Title

Date

Screenshot:

- ```
child: Column(
 mainAxisAlignment: MainAxisAlignment.center,
 crossAxisAlignment: CrossAxisAlignment.center,
 children: [
 Row(
 mainAxisAlignment: MainAxisAlignment.spaceBetween,
 children: [
```



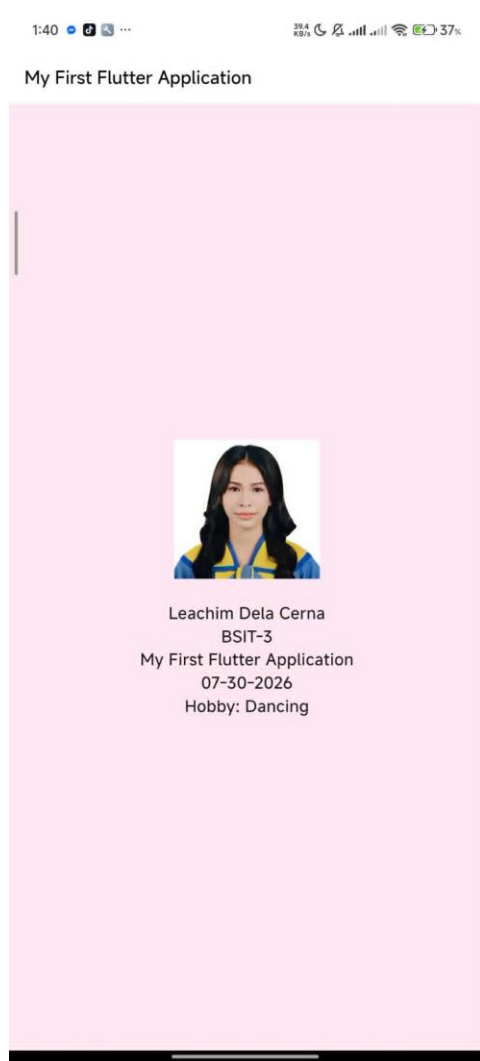
**Objective:** Add another individual widget to your `Column`.

Examples:

It must be its **own widget**, not additional text inside another `Text`.

- New widget in your Column
- Updated application

```
// HOBBY - NEW INDIVIDUAL WIDGET
const Text('Hobby: Dancing', textAlign: TextAlign.center),
```



## 🚩 Flag 4 – Put Two Things Side-by-Side

**Objective:** Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21      Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

## Column

└─ Text

└─ Text

└─ Text

└─ Text

└─ Text

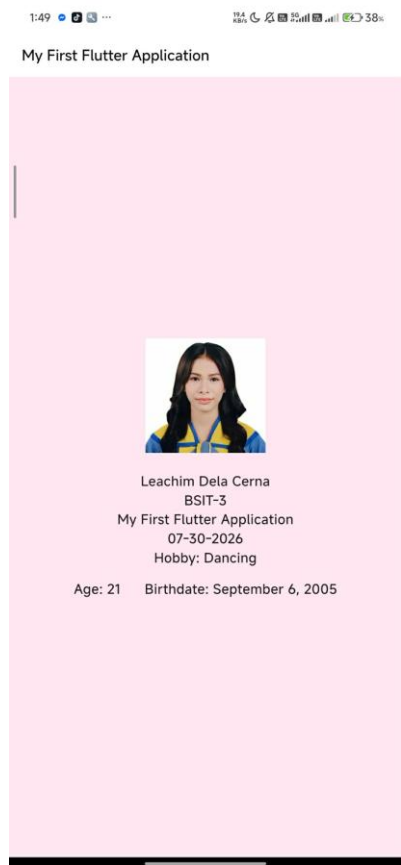
└─ Row

└─ Text

└─ Text

## Screenshot:

- Your widget structure/code
- The two items displayed side-by-side



```
// ROW - AGE AND BIRTHDATE
Row(
 mainAxisAlignment: MainAxisAlignment.center,
 children: [
 // AGE
 const Text('Age: 21', textAlign: TextAlign.center),

 const SizedBox(width: 30),

 // BIRTHDATE
 const Text(
 'Birthdate: September 6, 2005',
 textAlign: TextAlign.center,
), // Text
],
)
```

## 🚩 Flag 5 – Give It Some Space

**Objective:** Improve the spacing between your widgets.

Research:

`SizedBox ()`

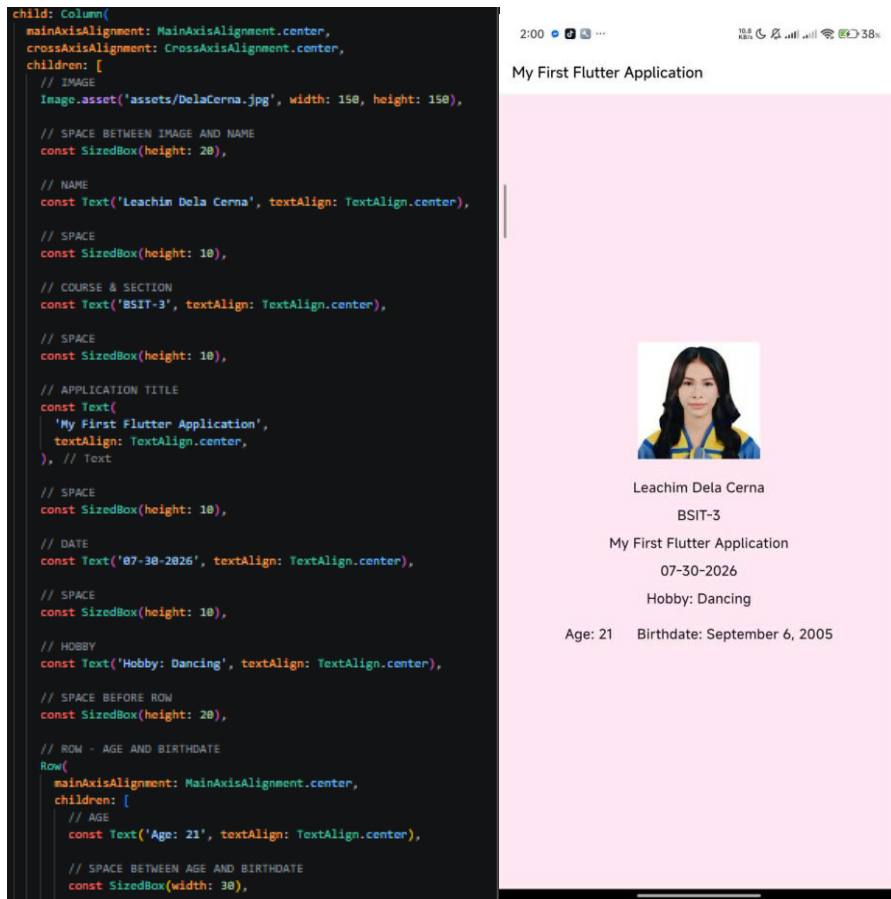
Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

**Challenge:** Make the card easier to read **without changing its colors or fonts yet**.

**Screenshot:**

- `SizedBox` usage
- Updated application



## 🚩 Flag 6 – Make It Look Better

**Objective:** Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

Apply different styling to at least **three** elements.

For example:

- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

**Screenshot:**

- Your `TextStyle` changes
- Updated application



```
// NAME - LARGER AND BOLD
const Text(
 "Leachim Dela Cerna",
 textAlign: TextAlign.center,
 style: TextStyle(
 fontSize: 26,
 fontWeight: FontWeight.bold,
 color: Colors.black,
), // TextStyle
), // Text

// SPACE
const SizedBox(height: 10),

// COURSE & SECTION - MEDIUM SIZE AND BOLD
const Text(
 "BSIT-3",
 textAlign: TextAlign.center,
 style: TextStyle(
 fontSize: 20,
 fontWeight: FontWeight.bold,
 color: Colors.black,
), // TextStyle
), // Text

// SPACE
const SizedBox(height: 10),

// APPLICATION TITLE - ITALIC
const Text(
 "My First Flutter Application",
 textAlign: TextAlign.center,
 style: TextStyle(
 fontSize: 18,
 fontStyle: FontStyle.italic,
 color: Colors.black,
), // TextStyle
), // Text

// SPACE
const SizedBox(height: 10),

// DATE - SPILLER
const Text(
 "07-30-2026",
 textAlign: TextAlign.center,
 style: TextStyle(fontSize: 14, color: Colors.black),
), // Text

// SPACE
const SizedBox(height: 10),

// HOBBY - STYLED
const Text(
 "Hobby: Dancing",
 textAlign: TextAlign.center,
 style: TextStyle(
 fontSize: 18,
 fontWeight: FontWeight.w500,
 color: Colors.black,
), // TextStyle
), // Text

// SPACE BEFORE ROW
const SizedBox(height: 20),

// ROW - AGE AND BIRTHDATE
Row(
 mainAxisAlignment: MainAxisAlignment.center,
 children: [
 // AGE
 const Text(
 "Age: 21",
 textAlign: TextAlign.center,
 style: TextStyle(fontSize: 16, color: Colors.black),
), // Text

 // SPACE BETWEEN AGE AND BIRTHDATE
 const SizedBox(width: 10),

 // BIRTHDATE
 const Text(
 "Birthdate: September 6, 2005",
 textAlign: TextAlign.center,
 style: TextStyle(fontSize: 16, color: Colors.black),
), // Text
],
)
```

## 🚩 Flag 7 – Improve the Layout

**Objective:** Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

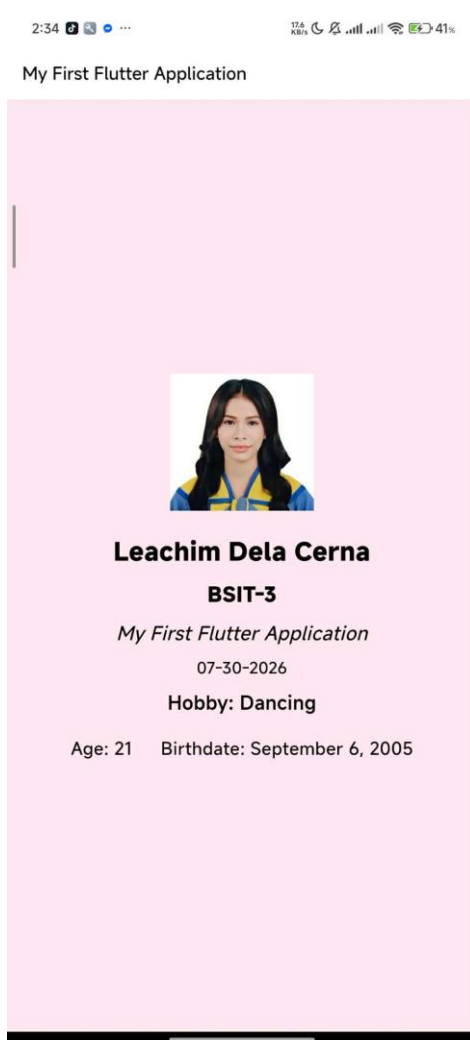
`mainAxisAlignment`

`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

**Screenshot:**

- The properties you changed
- Updated application



```
// ROW - AGE AND BIRTHDATE
Row(
 mainAxisAlignment: MainAxisAlignment.center,
 crossAxisAlignment: CrossAxisAlignment.center,
 children: [
 // AGE
 const Text(
 'Age: 21',
 textAlign: TextAlign.center,
 style: TextStyle(fontSize: 16, color: Colors.black),
), // Text
 // SPACE BETWEEN AGE AND BIRTHDATE
 const SizedBox(width: 30),
 // BIRTHDATE
 const Text(
 'Birthdate: September 6, 2005',
 textAlign: TextAlign.center,
 style: TextStyle(fontSize: 16, color: Colors.black),
), // Text
],
)
```

## Flag 8 – Create a Second Card

**Objective:** Apply what you learned to create another section.

Create a second `Card` containing something different from your profile.

For example:

**My Favorites**

Favorite Game

Favorite Food

Favorite Movie

The second card should also use:

- Individual `Text` widgets
- A `Column`
- At least one `Row`
- Spacing

**Screenshot:**

- Code for the second card
- Both cards displayed

```
// SPACE
const SizedBox(height: 15),

// FAVORITE MOVIE
const Text(
 'Favorite Movie',
 style: TextStyle(
 fontSize: 18,
 fontWeight: FontWeight.bold,
 color: Colors.black,
), // TextStyle
), // Text

const SizedBox(height: 5),

const Text(
 'Avengers: Endgame',
 style: TextStyle(fontSize: 16, color: Colors.black),
), // Text

// SPACE
const SizedBox(height: 28),

// ROW
Row(
 mainAxisAlignment: MainAxisAlignment.spaceEvenly,
 crossAxisAlignment: CrossAxisAlignment.center,
 children: [
 const Text(
 'Game: Mobile Legends',
 style: TextStyle(
 fontSize: 14,
 color: Colors.black,
), // TextStyle
), // Text

 const Text(
 'Food: Fried Chicken',
 style: TextStyle(
 fontSize: 14,
 color: Colors.black,
), // TextStyle
), // Text
],
),
```

## My First Flutter Application



## My First Flutter Application

## My First Flutter Application

07-30-2026

**Hobby:** Dancing

Age: 21    Birthdate: September 6, 2005

## My Favorites

### Favorite Game

## Mobile Legends

**Favorite Food**

Fried Chicken

**Favorite Movie**

Avengers: Endgame

Game: Mobile Legends    Food: Fried Chicken

## 🚩 Flag 9 – Stack the Cards

**Objective:** Arrange your two cards vertically.

You now have:

Profile Card



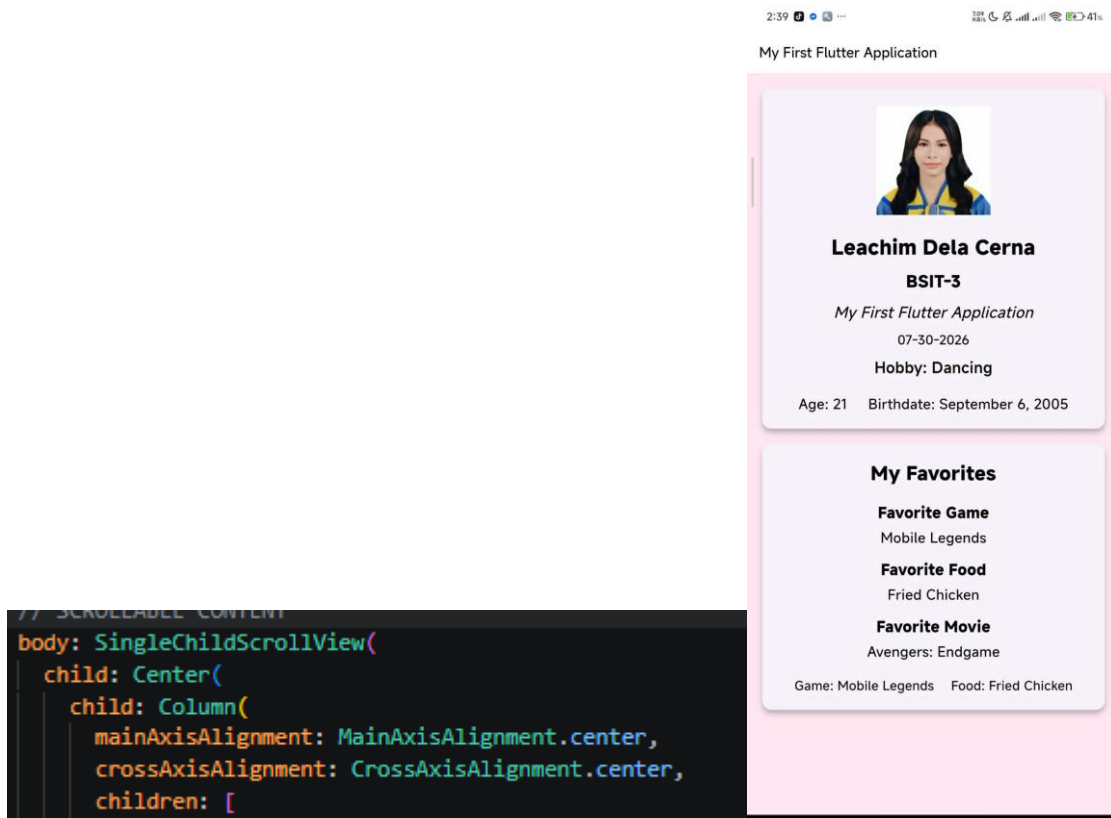
Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

**Screenshot:**

- The widget containing both Cards
- Both cards visible in the application



## 🚩 Flag 10 – Widget Discovery

**Objective:** Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

Examples:

- `Icon`
- `Divider`
- `CircleAvatar`
- `Chip`
- `ElevatedButton`

Add it somewhere appropriate in your application.

**Screenshot:**

- Your newly discovered widget in the code
- The widget visible in the application



3:03



7:31 KB/s



44%

## My First Flutter Application



### Leachim Dela Cerna

BSIT-3

*My First Flutter Application*

July 30, 2026

Hobby: Dancing

Age: 21

Birthdate: September 6, 2005

### ♥ My Favorites ♥

#### Favorite Game

Mobile Legends

#### Favorite Food

Fried Chicken

#### Favorite Movie

Avengers: Endgame

Game: Mobile Legends

Food: Fried Chicken

## 🚩 Flag 12 – Save Your Progress

**Objective:** Commit your completed Widget branch.

Follow proper standards of commit messages

**Screenshot:**

- Successful commit
- `feature/widgets` branch

```
Student@PC-03 MINGW64 C:/Users/Student/Desktop/IT314AB_DelaCerna/my_first_flutter_app (feature/widgets)
$ git branch
* feature/widgets
main

Student@PC-03 MINGW64 C:/Users/Student/Desktop/IT314AB_DelaCerna/my_first_flutter_app (feature/widgets)
$ git status
On branch feature/widgets
Your branch is up to date with 'origin/feature/widgets'.

Changes to be committed:
 (use "git restore --staged <file>..." to unstage)
 modified: lib/main.dart

Student@PC-03 MINGW64 C:/Users/Student/Desktop/IT314AB_DelaCerna/my_first_flutter_app (feature/widgets)
$ git add .

Student@PC-03 MINGW64 C:/Users/Student/Desktop/IT314AB_DelaCerna/my_first_flutter_app (feature/widgets)
$ git commit -m "feat: submit Capture The Flag activity 2"
[feature/widgets 5462709] feat: submit Capture The Flag activity 2
1 file changed, 223 insertions(+), 73 deletions(-)

Student@PC-03 MINGW64 C:/Users/Student/Desktop/IT314AB_DelaCerna/my_first_flutter_app (feature/widgets)
$ git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 1.66 KiB | 1.66 MiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/chm0-boop/IT314AB_DelaCerna.git
 dbc2fbd..5462709 feature/widgets -> feature/widgets
```